UNIT- IV

Sustainable Design Approaches

Concern for Environment and Sustainability in Design, Case Studies to understand good Design for Environment (DFE) Decisions; Design Considerations in the five stages of the Product Life Cycle.

Eco-design: how to manufacture sustainable products to satisfy consumers









IBERDROLA

The benefits of ECO-DESIGN

HIGHER QUALITY PRODUCTS

Eco-design products are **more versatile** and are manufactured with longer-lasting materials.

MORE EFFICIENT PRODUCTION

They **save energy** and require fewer natural resources and raw materials.

FEWER EMISSIONS

They consume less energy during transport for lower **CO_2 emissions.**

MORE SUSTAINABLE INDUSTRIES

Companies benefit from **innovation** and become more committed to the environment.

HAPPIER CONSUMERS

Consumers' needs are met with **more attractive products** that satisfy an increasingly demanding public.

MARKET DIFFERENTIATION

Sustainable products have **added value** hat gives them an edge over competitors.

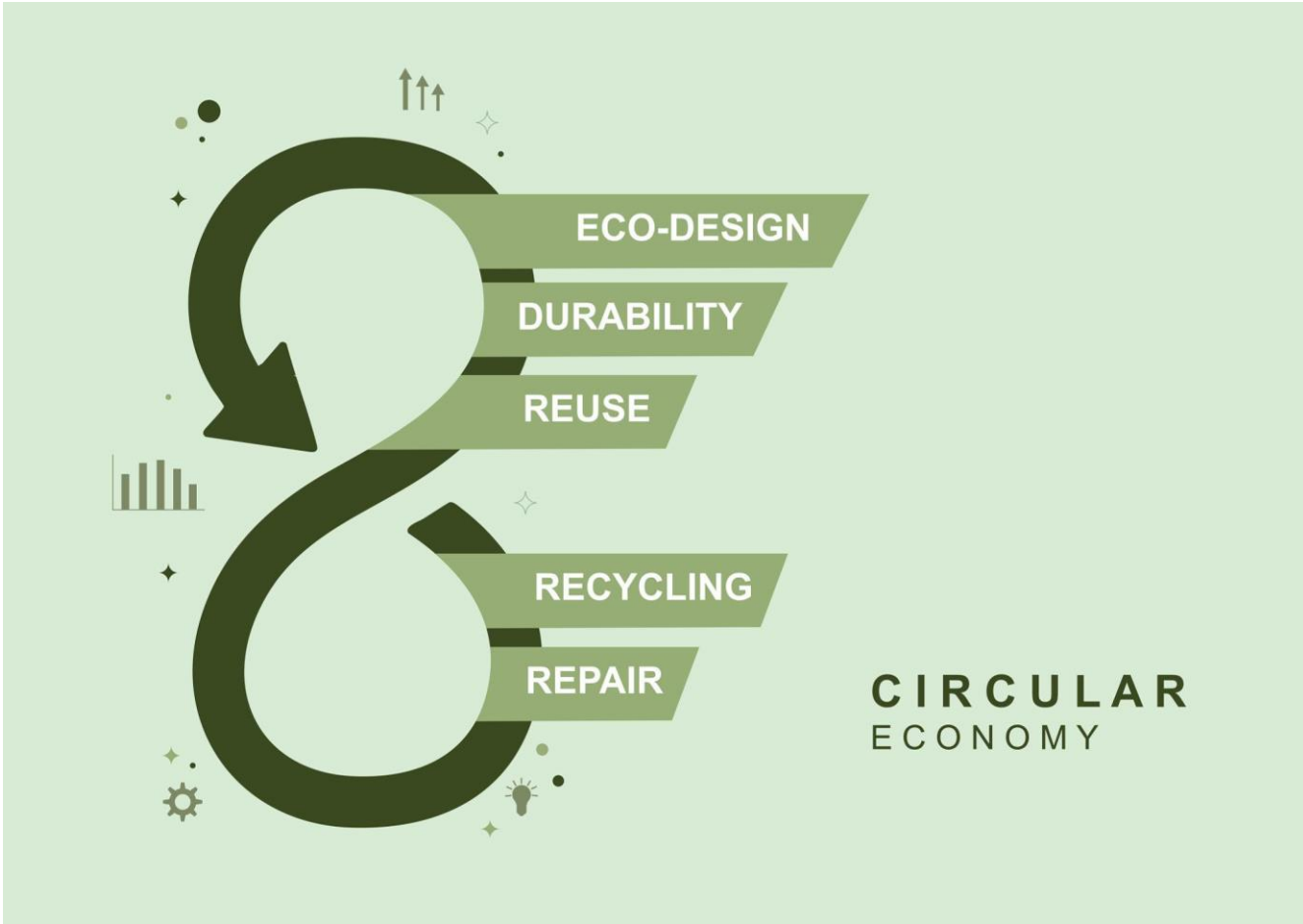
Source Euro-Funding.

https://www.iberdrola.com/

Sustainable Design is the approach of designing products and services that **minimize environmental impact**, conserve resources, and promote long-term ecological balance.

Objectives

- Reduce pollution
- Conserve energy and materials
- Promote recyclability
- Extend product life
- Reduce carbon footprint



Why Environmental Concern is Important

- Climate change
- Resource depletion
- Waste generation
- Increasing environmental regulations
- Consumer demand for eco-friendly products

Principles of Sustainable Design

1. **Life Cycle Thinking** – Consider environmental impact from raw material to disposal
2. **Reduce, Reuse, Recycle (3R)**
3. **Energy Efficiency**
4. **Material Optimization**
5. **Renewable Resources Usage**
6. **Low Carbon Emissions**

Sustainable Design Strategies

- Lightweight materials
- Modular design
- Biodegradable materials
- Green manufacturing
- Renewable energy integration


Resources
Material processing
Product Manufacturing
Distribution
Product Use
End of Life
Sustainable Design

Design for Environment (DFE)

Design for Environment (DFE) is a design approach that integrates **environmental considerations into product and service design** to minimize environmental impact throughout the product life cycle.

It aims to:

- Reduce pollution
- Minimize resource consumption
- Improve recyclability
- Promote sustainable product development

DFE is also called **Eco-design** or **Green Design**.

The main objectives of DFE are:

- Reduce environmental impact of products
- Use eco-friendly materials
- Improve energy efficiency
- Minimize waste generation
- Promote reuse and recycling
- Extend product life span

Principles of Design for Environment

1. Material Selection

Choose materials that are:

- Non-toxic
- Recyclable
- Renewable
- Lightweight

Example: Using biodegradable packaging instead of plastic.

2. Energy Efficiency

Products should consume **less energy during production and use**.

Example: LED bulbs consume much less energy compared to incandescent bulbs.

3. Waste Reduction

Design products to minimize waste during manufacturing.

Example: Precision manufacturing techniques reduce scrap materials.

5. Product Durability

Design products with **longer life spans** to reduce frequent replacement.

Example: Durable appliances with repairable components.

4. Design for Disassembly

Products should be designed so components can be **easily separated and recycled**.

Example: Modular smartphones that allow easy replacement of parts.

Advantages of Design for Environment

- Reduced environmental pollution
- Improved product sustainability
- Reduced operational costs
- Better compliance with environmental laws
- Increased customer acceptance

Challenges in Implementing DFE

- High initial design cost
- Limited availability of eco-friendly materials
- Lack of awareness in industries
- Technological limitations

Case Studies for DFE

Electric Vehicles (EVs)

Problem

Air pollution and fossil fuel dependence.

DFE Decision

- Use electric powertrain
- Regenerative braking
- Lightweight materials

Result

- Zero tailpipe emissions
- Reduced fuel dependency

Sustainable Packaging

Problem

Plastic waste accumulation.

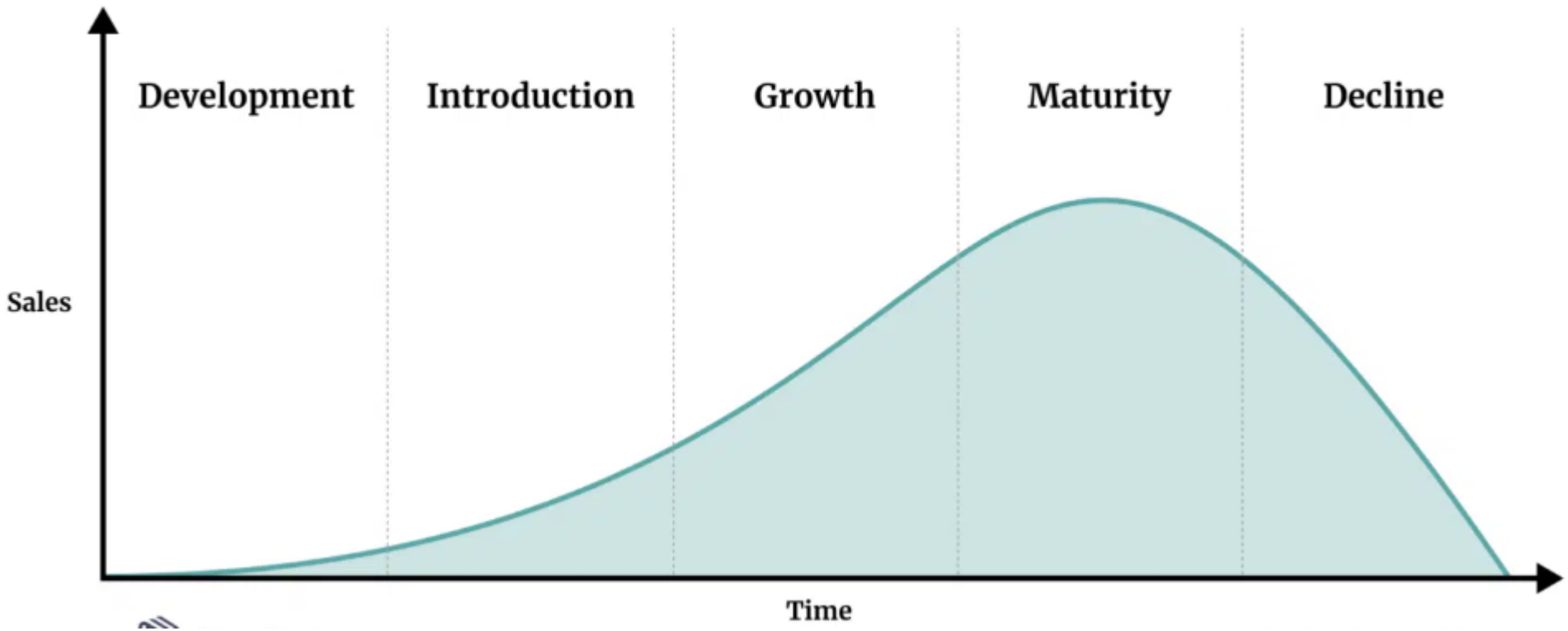
DFE Decision

- Biodegradable packaging
- Recyclable paper materials
- Reduced packaging size

Result

- Reduced landfill waste
- Lower transportation emissions

The Product Life Cycle



The **Product Life Cycle (PLC)** describes the stages a product goes through from **initial development to decline in the market**. Understanding these stages helps companies plan marketing, design improvements, and sustainability strategies.

1 Development Stage (Product Development)

- Product idea is researched and designed.
- Prototypes are created and tested.
- High investment costs with **no sales revenue**.
- Focus on **research, innovation, and testing**.

Example: Development of a new electric vehicle model before launch.

2 Introduction Stage

- Product is launched into the market.
- Sales begin slowly because awareness is low.
- High promotion and marketing expenses.
- Limited competition.

Example: First launch of a new smartphone model.

3 Growth Stage

- Rapid increase in sales and market acceptance.
- Competitors start entering the market.
- Product improvements and feature additions occur.
- Profits increase significantly.

Example: Increasing adoption of smartwatches after initial launch.

4 Maturity Stage

- Market becomes saturated.
- Sales growth slows or stabilizes.
- Competition becomes intense.
- Companies focus on **differentiation and cost reduction**.

Example: Conventional LED televisions in the consumer market.

5 Decline Stage

- Sales decrease due to new technologies or changing customer preferences.
- Profits decline.
- Companies may discontinue or redesign the product.

Example: DVD players declining due to streaming services.

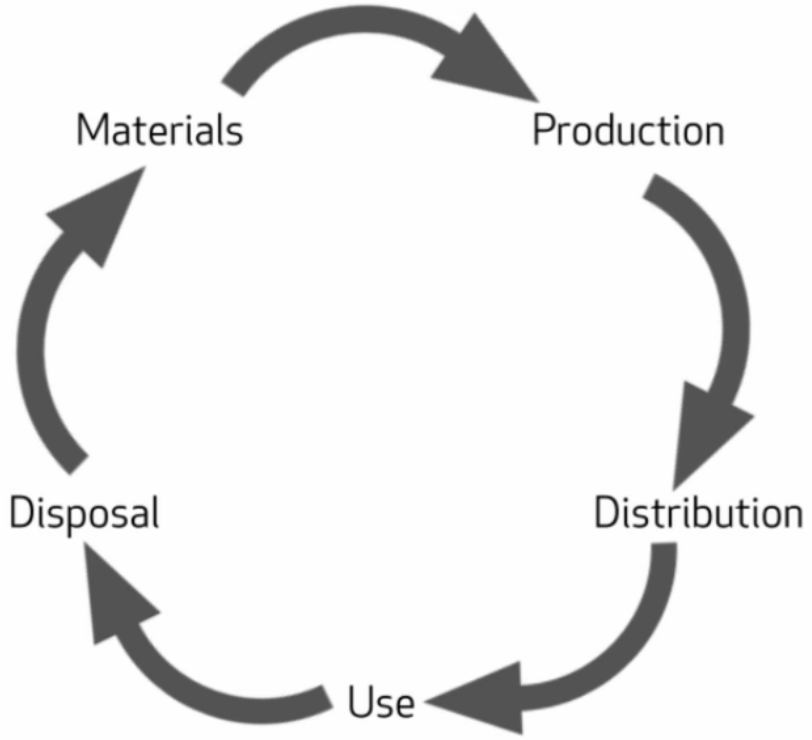
Design Considerations in the five stages of the Product Life Cycle.

The **Product Life Cycle (PLC)** represents the stages a product passes through from **raw material extraction to disposal or recycling**.

In sustainable design, engineers must consider **environmental impact at each stage** to reduce waste, pollution, and energy consumption.

The five major stages are:

1. Raw Material Extraction
2. Manufacturing
3. Distribution
4. Usage
5. End-of-Life (Disposal / Recycling)



Stage 1: Raw Material Extraction

Design Considerations

- Use **renewable or recyclable materials**
- Reduce material usage through **lightweight design**
- Avoid **toxic or hazardous substances**
- Prefer **locally available materials** to reduce transportation emissions
- Choose materials with **low environmental impact**

Example

Using **recycled aluminum** instead of newly mined aluminum in product manufacturing.

Stage 2: Manufacturing

Design Considerations

- Use **energy-efficient manufacturing processes**
- Reduce **industrial waste and emissions**
- Minimize **water usage**
- Implement **cleaner production technologies**
- Optimize production to reduce material scrap

Example

Automobile companies using **robotic precision manufacturing** to minimize material waste.

Stage 3: Distribution

Design Considerations

- Design **compact products and packaging**
- Use **lightweight packaging materials**
- Optimize **transportation logistics**
- Reduce **fuel consumption and emissions**
- Encourage **eco-friendly packaging**

Example

Flat-pack furniture (like IKEA) reduces transportation volume and fuel consumption.

Stage 4: Usage

Design Considerations

- Improve **energy efficiency during operation**
- Design products with **long lifespan**
- Enable **easy maintenance and repair**
- Reduce **resource consumption during use**
- Ensure **user safety and convenience**

Example

Energy-efficient appliances such as **LED bulbs or inverter air conditioners**.

Stage 5: End-of-Life (Disposal / Recycling)

Design Considerations

- Design products for **easy disassembly**
- Use **recyclable or biodegradable materials**
- Promote **reuse and remanufacturing**
- Implement **take-back programs**
- Reduce landfill waste

Example

Smartphones designed with **recyclable components and modular parts**.


BODY MOISTURIZER
SWEET BASIL
ECO
Please Recycle This Bag





Bamboo Water Bottles Made By An Indian Student As An Alterna Plastic Is Brilliant



https://noonecares.me/